STUDENT PERFORMANCE

Educational Equity and Insights Report

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Register Number: 26BSC106

Course: B.Sc. Computer Science (AI & DS)

Dataset: StudentsPerformance.csv

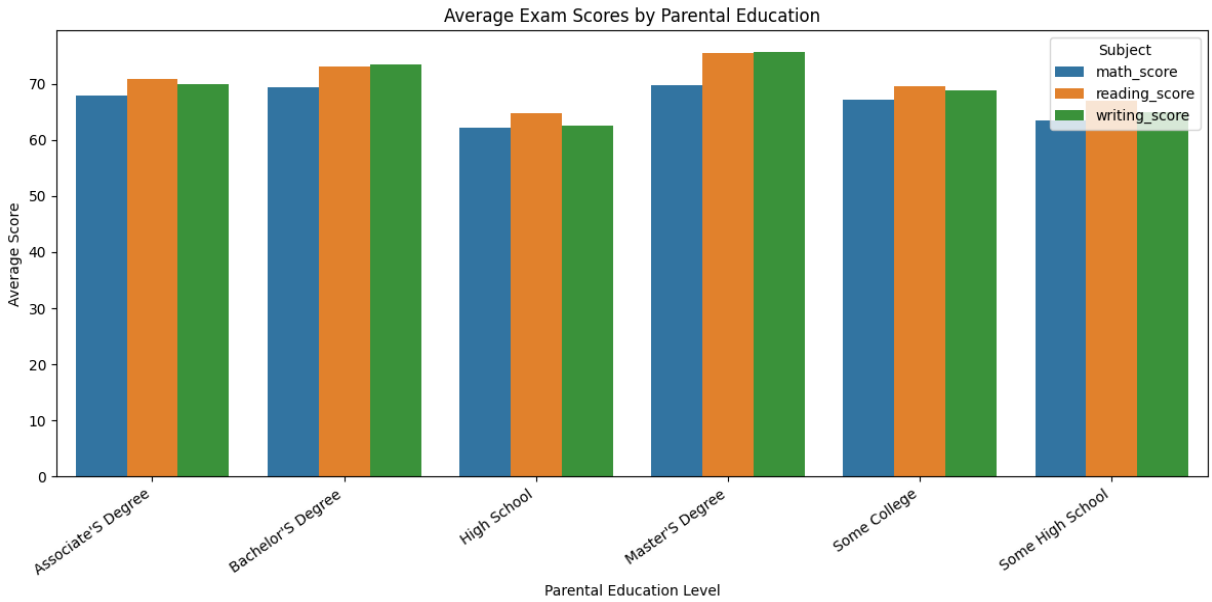
Tools Used: Google Colab, Python, Pandas, NumPy, Matplotlib and Seaborn

1. Introduction

- This report examines the StudentsPerformance.csv dataset, which contains information about 1,000 students and their examination scores in Mathematics, Reading and Writing. The dataset includes five categorical fields—gender, race/ethnicity, parental level of education, lunch category and test preparation course—along with the three subject scores.
- The purpose of the analysis is to explore how average academic scores vary across selected student groups, compare performance according to test preparation status, and examine the relationships among the three subject scores. The report focuses on describing patterns in the available data rather than making claims about the causes of student performance.
- The notebook reports 1,000 rows and 8 original columns. All eight columns contain 1,000 non-null values, and the initial duplicate check found no duplicate rows. The overall mean scores were 66.09 in Mathematics, 69.17 in Reading and 68.05 in Writing. The analysis was carried out in Python using Pandas and NumPy for data handling and Matplotlib and Seaborn for visualisation.

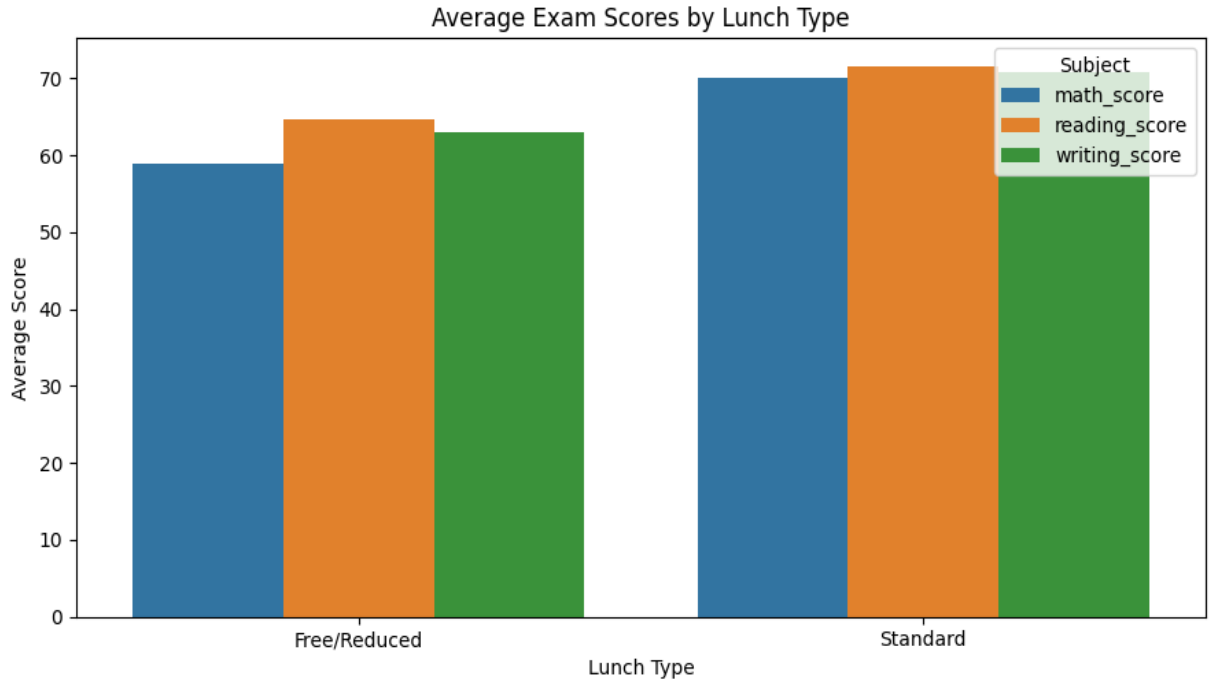
2. Analysis of Parental Education

- The grouped bar chart compares the mean Mathematics, Reading and Writing scores across six parental education categories. The highest averages in Reading and Writing appear in the Master's Degree group (75.37 and 75.68 respectively). This group also has a Mathematics average of 69.75. The Bachelor's Degree group records averages of 69.39 in Mathematics, 73.00 in Reading and 73.38 in Writing.
- The High School category has the lowest mean scores among the six groups: 62.14 in Mathematics, 64.70 in Reading and 62.45 in Writing. Students in the Some High School category also have comparatively lower averages than several other groups. Across the categories, Reading and Writing averages are generally higher than Mathematics averages.
- These results show an association between the recorded parental education categories and students' average scores in this dataset. They do not show that parental education alone causes a particular academic outcome; other student, family, school and resource-related factors may also be relevant.



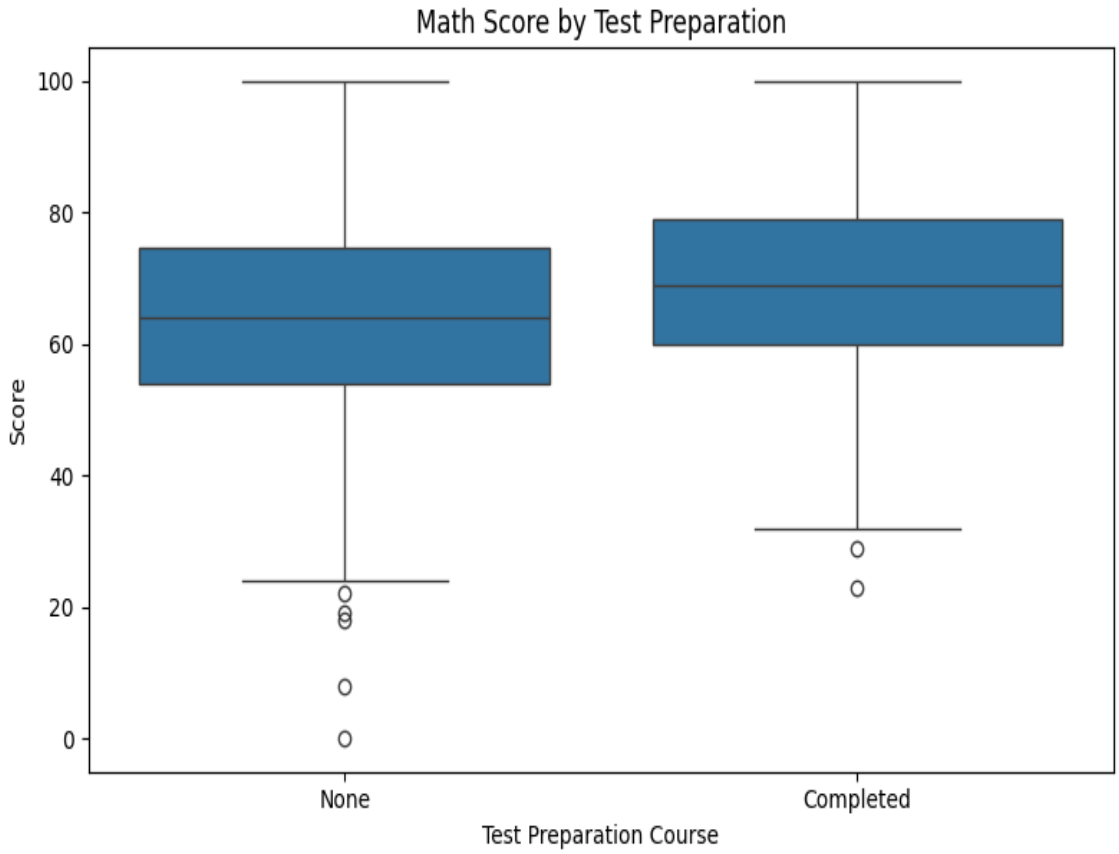
3. Analysis of Lunch Type

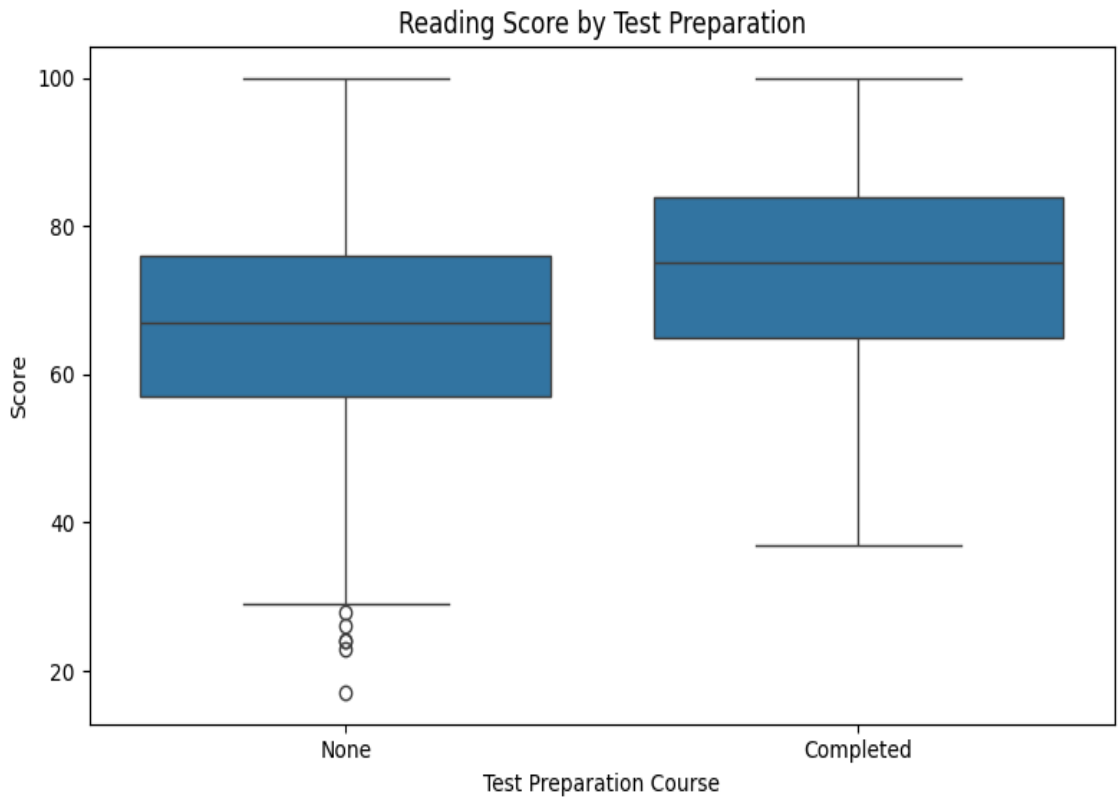
- The grouped bar chart compares average Mathematics, Reading and Writing scores for the Standard and Free/Reduced lunch categories. In the notebook's visualisation, the Standard lunch category shows higher average scores across the three subjects than the Free/Reduced category. The same broad pattern is visible across Mathematics, Reading and Writing.
- Lunch category may provide a limited indicator of differences in students' socioeconomic circumstances or access to resources. The observed score differences should therefore be considered as group-level patterns that may warrant further investigation. Lunch category does not measure a student's complete financial or home situation, and the chart cannot establish that lunch type causes differences in examination performance.

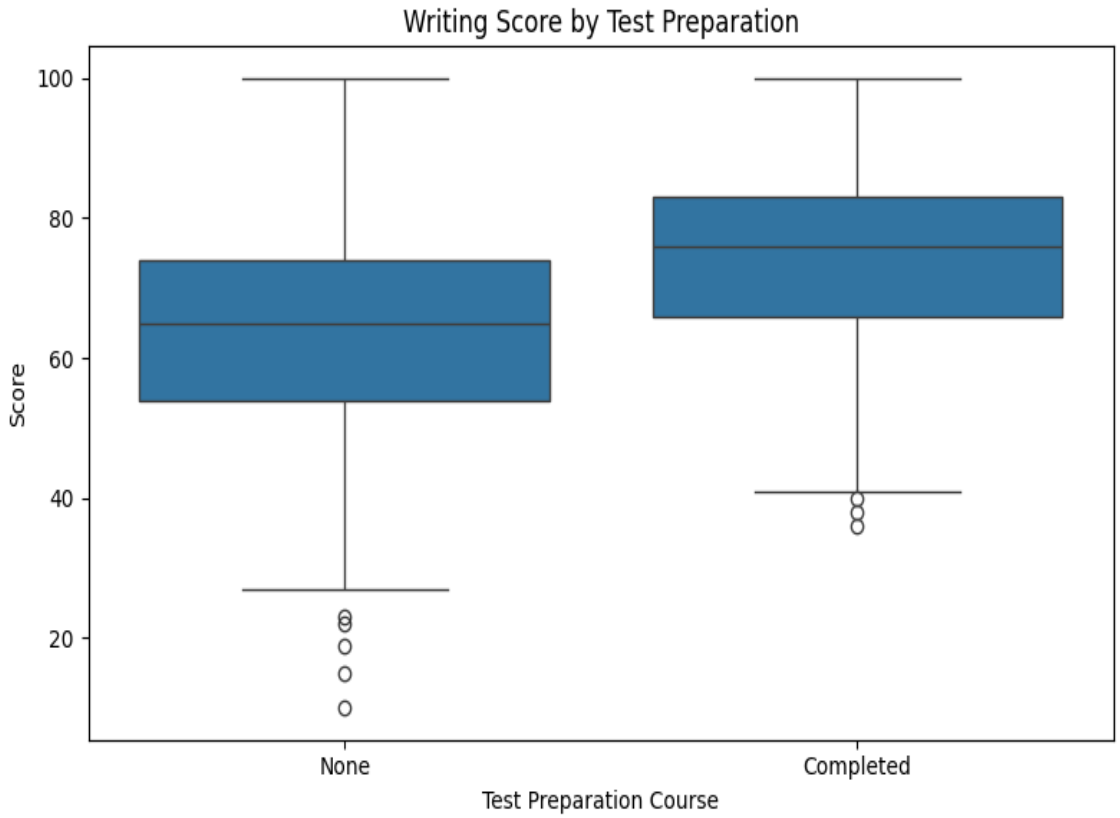


4. Test Preparation Analysis

- The box plots compare subject-score distributions for students who completed a test preparation course and students recorded as having no completed course. In all three subjects, the completed group has a higher median and a higher mean score. The median scores for completed preparation were 69 in Mathematics, 75 in Reading and 76 in Writing. For the group with no completed preparation, the corresponding medians were 64, 67 and 65.
- The mean scores also show this pattern. Students who completed preparation averaged 69.70 in Mathematics, 73.89 in Reading and 74.42 in Writing. Students with no completed course averaged 64.08, 66.53 and 64.50 respectively. The standard deviations were slightly lower for the completed group in each subject: 14.44, 13.64 and 13.38, compared with 15.19, 14.46 and 15.00 for the group with no completed course. This indicates somewhat less score variation in the completed group in this dataset.
- The box plots also show the spread of scores and potential outlying observations. The notebook's separate IQR check identified 8 potential Mathematics outliers, 6 Reading outliers and 5 Writing outliers across the full dataset. These counts are not separated by preparation category in the recorded output. The group differences are descriptive and do not prove that the preparation course itself caused higher scores.

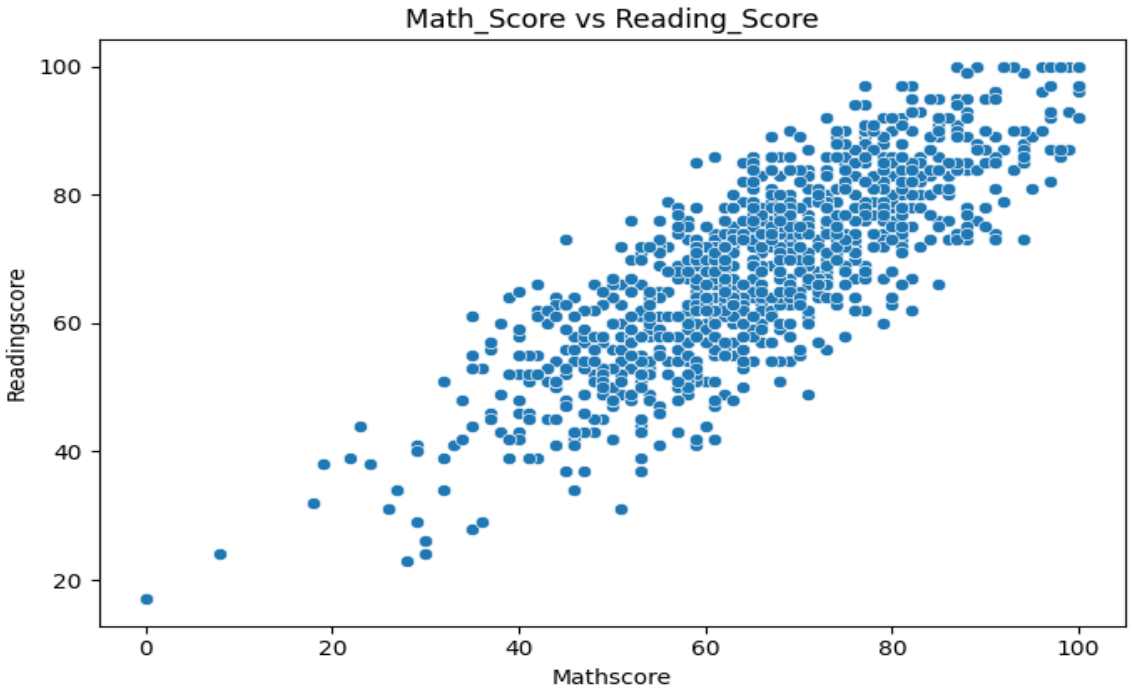


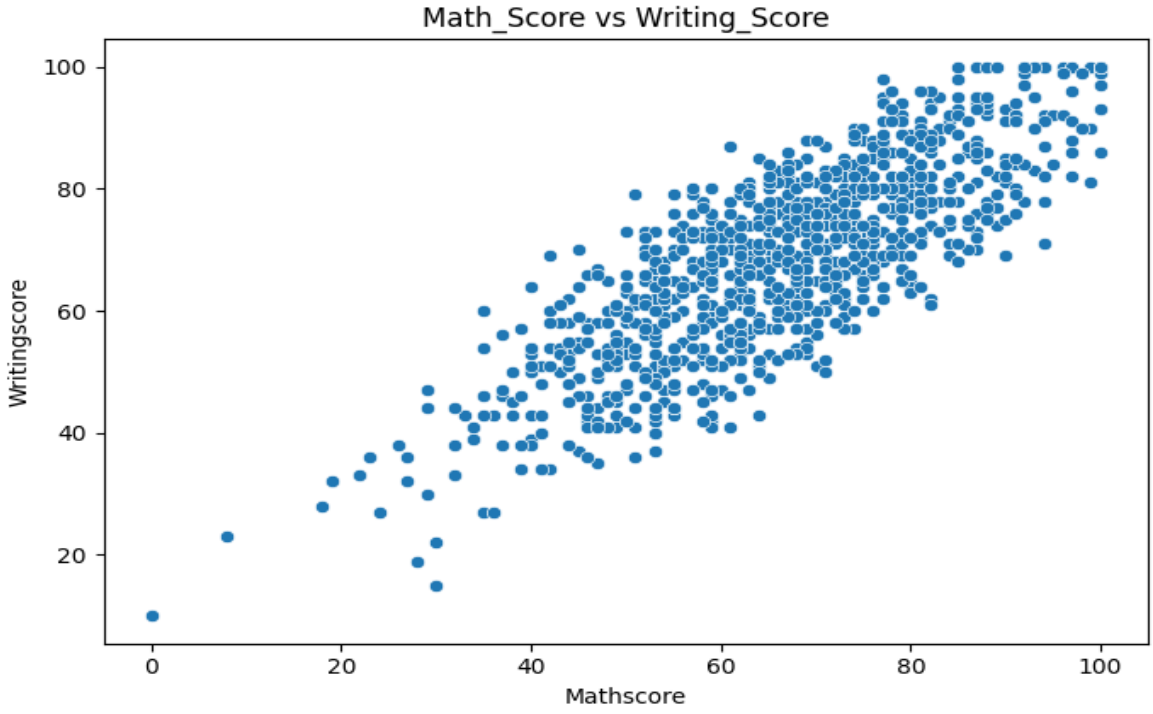

Reading Score by Test Preparation
Score
100
80
60
40
20
None
Completed
Test Preparation Course

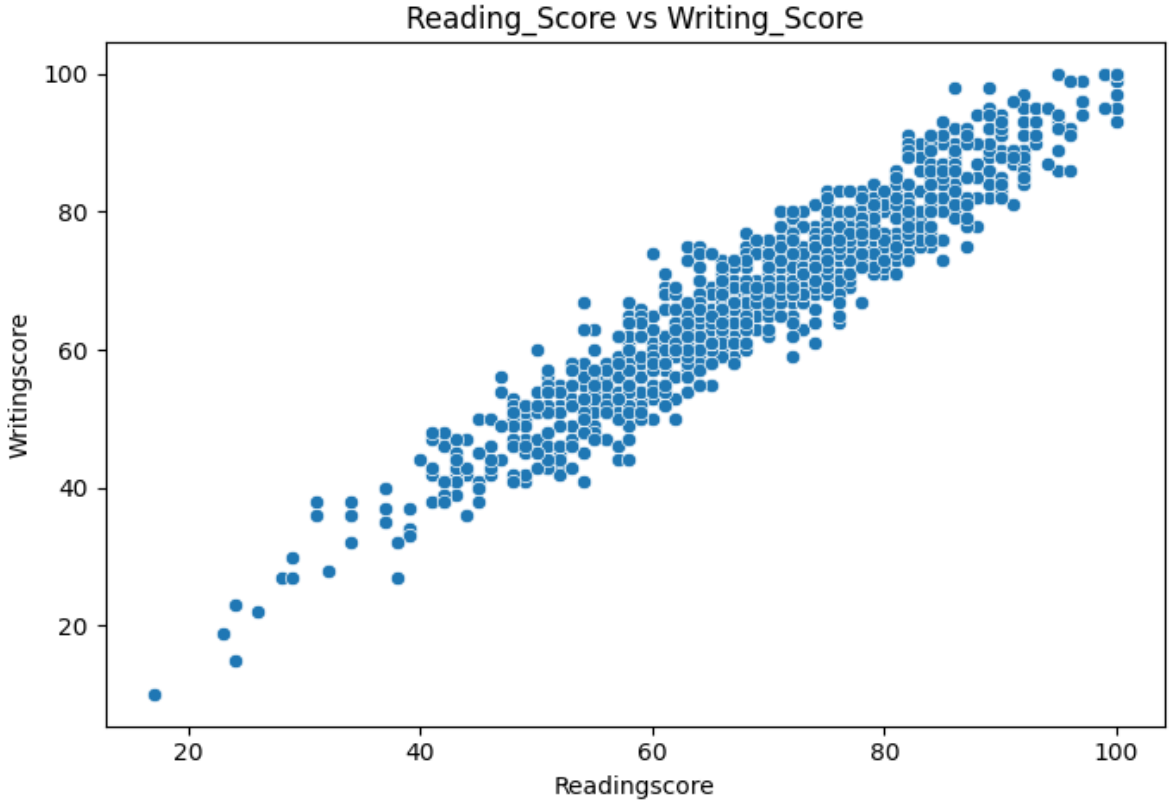

Writing Score by Test Preparation
Score
100
80
60
40
20
None
Completed
Test Preparation Course

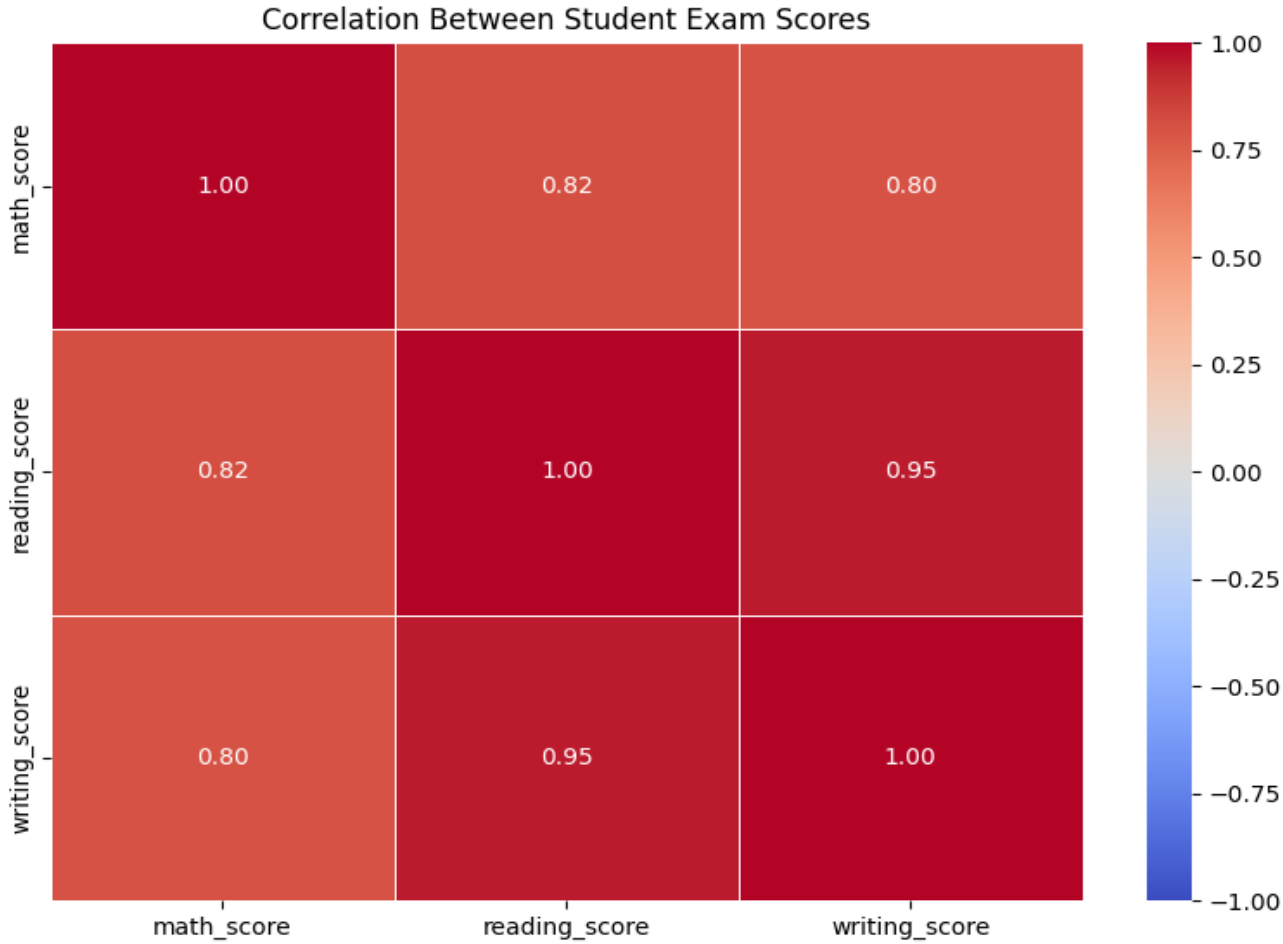
5. Subject Correlation Analysis

- Scatter plots are used to examine how scores in one subject vary alongside scores in another. The points generally form upward patterns, indicating positive relationships: students with higher scores in one subject often also have higher scores in another.
- The correlation matrix quantifies these relationships. Reading and Writing have the strongest positive correlation (r = 0.9546). Mathematics and Reading have a correlation of 0.8176, while Mathematics and Writing have a correlation of 0.8026. All three coefficients are positive, and the Reading–Writing relationship is the strongest among the subject pairs examined.
- A correlation describes the strength and direction of a linear relationship; it does not demonstrate that performance in one subject causes performance in another.






Reading_Score vs Writing_Score
Writingscore
Readingscore
100
80
60
40
20
20
40
60
80
100


Correlation Between Student Exam Scores
math_score
reading_score
writing_score
1.00
0.82
0.80
0.82
1.00
0.95
0.80
0.95
1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

6. Educational Equity Recommendations

The findings can help identify areas for further attention. The following recommendations are practical starting points and should be evaluated with additional evidence before being adopted as formal policy:

- Provide additional academic support where lower average performance is observed, including focused help in Mathematics and support for groups that show lower averages.
- Improve access to test preparation resources so that students can participate regardless of their financial circumstances, location or access to outside support.
- Offer targeted support in subjects with weaker average performance. In this dataset, Mathematics has the lowest overall mean score (66.09).
- Monitor outcomes across demographic and background groups over time to identify persistent performance gaps and changes in those gaps.
- Evaluate any intervention using additional data, including participation, student progress and contextual information, before making policy decisions.

7. Limitations

- This report is based on the variables and records available in the StudentsPerformance.csv dataset. The analysis is descriptive and does not include every possible factor that may influence academic performance, such as school resources, attendance, teaching methods, home learning conditions or access to educational support.
- Differences in group averages and correlations do not establish causation. For example, the higher scores observed in the Standard lunch group cannot be attributed to lunch category itself. Similarly, the higher scores among students who completed test preparation do not prove that course completion alone produced the difference. Other factors may influence both participation and scores.
- The notebook provides numerical group summaries for parental education and test preparation, but it does not print a numerical lunch-group summary table. Therefore, the lunch comparison in this report is described from the saved chart rather than presented as exact group means. Further analysis with additional contextual variables and suitable statistical methods would be needed for stronger conclusions.

8. Conclusion

The analysis of 1,000 student records shows differences in average scores across parental education and lunch categories. The Master's Degree parental education group has the highest Reading and Writing averages, while the High School group has the lowest averages across the three subjects. The Standard lunch category also shows higher average scores than the Free/Reduced category in the saved grouped bar chart.Students who completed test preparation have higher mean and median scores in Mathematics, Reading and Writing, with slightly lower score variation than students who did not complete the course. The strongest relationship among subject scores is between Reading and Writing ($r = 0.9546$).Overall, the results highlight the value of accessible test preparation, targeted subject support and continued monitoring of educational outcomes. Further analysis using additional student and school context, and evaluation of carefully designed interventions, would help determine which forms of support are most effective and equitable.